

10. (2 points)

An actuary is reviewing the diagnostic results of a bootstrapping model.

a. (1 point)

Construct and label a plot of residuals that would indicate a calendar year trend due to inflation which is not reflected in the model, and describe how the residual pattern shows inflation.

b. (1 point)

Construct and label a plot of residuals that would indicate heteroscedasticity in development ages 36 and 48 compared to development ages 12 and 24, and describe how the residual pattern shows heteroscedasticity.

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EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

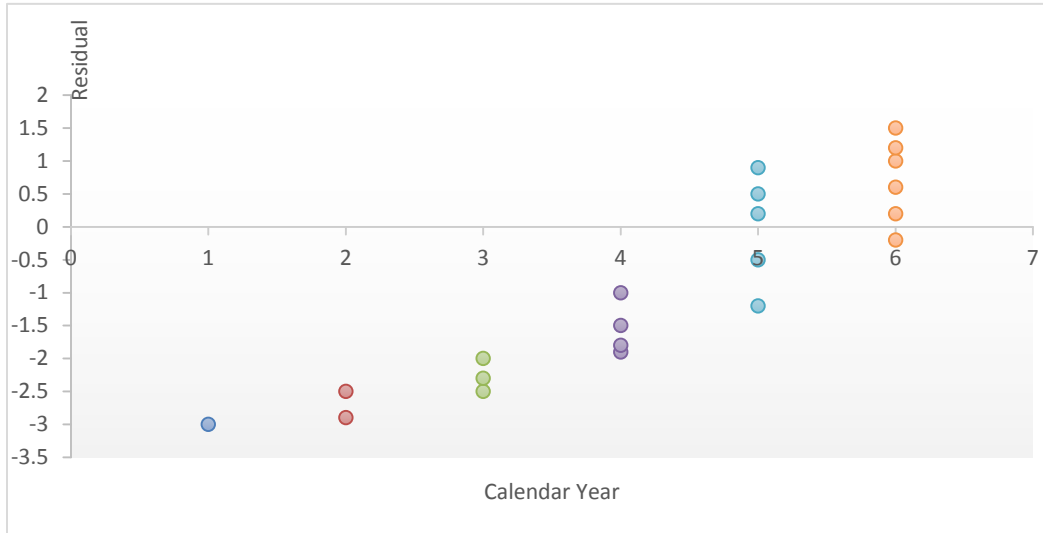
QUESTION 10

TOTAL POINT VALUE: 2

LEARNING OBJECTIVE(S): A9

SAMPLE ANSWERS

Part a: 1 point



Sample 1

We are systematically over-predicting at early calendar years and under-predicting at later calendar years, implying there is an increasing (i.e. inflation) effect for which our model is not accounting.

Sample 2

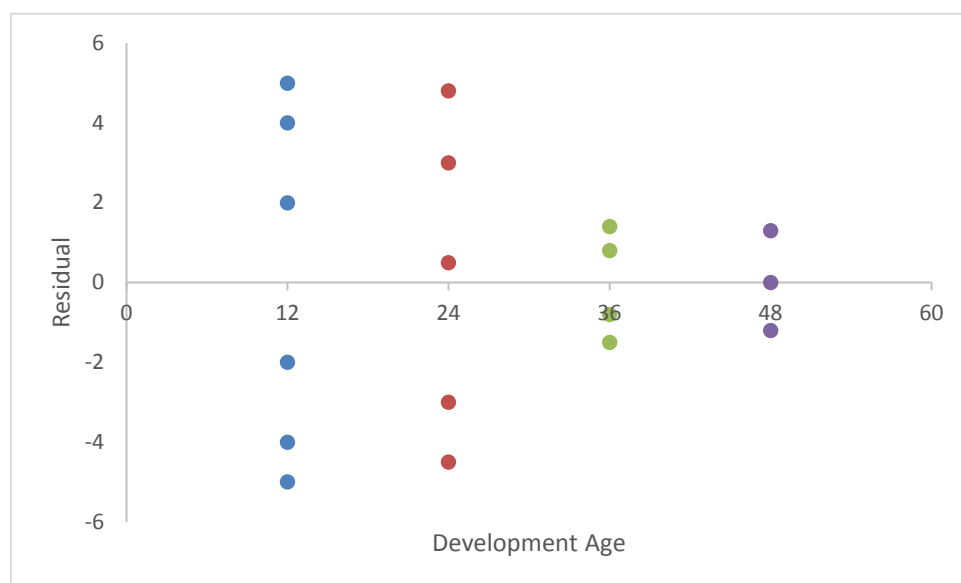
The model overstates losses for older CYs and underestimates losses for recent Cy's indicating a trend (i.e. inflation) not reflected in the model.

Sample 3

Since there is clearly an increasing pattern in the residuals from one CY to the next. This means the expected is overstated in early Cys and understated in recent Cys. This is a sign of CY trend due to inflation.

Part b: 1 point

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Sample 1

It shows heteroscedasticity because the variance is not constant across development periods; It is clearly higher (or lower) for 12 & 24 than it is for 36 & 48.

Sample 2

The residual show more (or less) spread at 12 and 24 month compared to 36 and 48 month.

Sample 3

Residuals for ages 12 and 24 are more wider spread than those at age 36 and 48 which means residuals do not have constant variance → heteroscedasticity.

EXAMINER'S REPORT

Candidates were expected to demonstrate the knowledge of testing assumptions underlying the bootstrap model.

Part a

Candidates were expected to construct a plot with labeled axis, zero-line and residuals at each calendar year showing upward trend (downward trend is acceptable if noted there was a deflation). Candidates were expected to describe how the upward trend indicates inflation not captured by the model.

Common errors include:

- Forgetting to label zero line.
- Providing an inappropriate pattern of number of residuals at each calendar year (the number of residuals should be increasing along the CY axis).
- Providing a description of the plot instead of reasoning how the trend indicates missing inflation captured by the model

Part b

Candidates were expected to construct a plot with labeled axis, zero-line and residuals at development age 36 and 48 showing similar dispersion but different with other ages. Candidates

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were expected to briefly explain how the residuals pattern demonstrated heteroscedasticity.

Common errors include:

- Forgetting to label the zero line.
- Providing an inappropriate number pattern of residuals at each development age (the number of residuals should be decreasing along the development age axis).
- Not plotting residuals at all required ages (12, 24, 36 and 48).
- Misunderstanding the variance concept and how variance is manifested in the plot.